

A Mechanical Route to G : Matching Dark–Matter Field Pressure to the Einstein–Hilbert Coupling

Robin Victor Skavberg*
Independent Researcher
Calgary, Alberta, Canada

January 26, 2026

Abstract

We develop a mechanical route to Newton’s gravitational coupling G by modeling the vacuum as a homogeneous, barotropic, irrotational quantum condensate. In the long-wavelength regime, the condensate’s field pressure (bulk modulus) fixes the mechanical stiffness to $K = \rho_m c^2$ when the signal speed is identified with c . We then compute the one-loop effective action of the medium and show that an Einstein–Hilbert (EH) term $\int \sqrt{-g} R$ is induced with a calculable coefficient. Matching this induced geometric stiffness to the condensate’s mechanical stiffness yields $G = c^2/(8\pi\rho_m)$, providing a consistent IR link between the medium and the observed gravitational coupling. Independently, we verify the identity $G = \Lambda c^4/(8\pi u_\nu)$ with Λ as the cosmological constant, furnishing an observational consistency check independent of microscopic details.

In a zero-sum balanced universe, G emerges as the necessary proportionality constant required to maintain the equilibrium between the vacuum’s mechanical stiffness and the observed geometric curvature.

1 Introduction

Analogue-gravity studies show that hydrodynamic media can reproduce relativistic kinematics via an effective metric,^{1–3} suggesting a conservative route to relate gravitational dynamics to mechanical properties. In this work, we develop a mechanical route to G by modeling the vacuum as a quantum condensate. Identifying the signal speed with c fixes the field pressure to $K = \rho_m c^2$, which we match to the geometric stiffness multiplying curvature in the Einstein–Hilbert (EH) action.^{4,5}

2 Assumptions and Regime of Validity

We assume: (i) long-wavelength hydrodynamics ($k\xi \ll 1$); (ii) homogeneous, barotropic background with mass density ρ_m ; (iii) phase coherence; (iv) identification of acoustic and electromagnetic null cones in the IR ($c_s = c$).

3 Hydrodynamic Framework

At the microscopic level, we take a complex scalar condensate described by the Gross–Pitaevskii (GP) equation:^{6,7}

$$i\hbar \partial_t \Psi = \left[-\frac{\hbar^2}{2m} \nabla^2 + V_{\text{ext}} + g|\Psi|^2 \right] \Psi. \quad (1)$$

*ORCID: [0009-0003-9126-6196](https://orcid.org/0009-0003-9126-6196)

Writing $\Psi = \sqrt{n} e^{i\theta}$ gives hydrodynamics for number density n and velocity $\mathbf{v} = (\hbar/m)\nabla\theta$. Linear perturbations propagate on an acoustic metric $ds^2 \propto -c_s^2 dt^2 + d\mathbf{x}^2$.

4 Mechanical Stiffness and Geometric Coupling

4.1 Bulk Modulus from the Equation of State

For the GP fluid, the pressure is $P = \frac{1}{2}gn^2$. We define $\rho_m \equiv mn$ as the mass density of the condensate (where m is the constituent mass and n is the number density). The sound speed is determined by the derivative of pressure with respect to the mass density:⁸

$$c_s^2 = \frac{dP}{d\rho_m} = \frac{d(\frac{1}{2}gn^2)}{d(mn)} = \frac{gn}{m}. \quad (2)$$

The bulk modulus (mechanical stiffness) is $\beta \equiv \rho_m \frac{dP}{d\rho_m}$. Setting $c_s = c$ yields:

$$K \equiv \beta = \rho_m c^2. \quad (3)$$

4.2 Matching to Einstein's Coupling

The EH action has the prefactor $c^4/(16\pi G)$, corresponding to a geometric stiffness $c^4/(8\pi G)$. Equating mechanical and geometric stiffnesses gives:

$$\rho_m c^2 = \frac{c^4}{8\pi G} \implies G = \frac{c^2}{8\pi \rho_m}. \quad (4)$$

4.3 Verification and Significance

The identity $G = \frac{\Lambda c^4}{8\pi u_\nu}$ provides the empirical bridge between field pressure and gravitation. Using current cosmological data,¹¹ the vacuum energy density $u_\nu \approx 5.3 \times 10^{-10}$ Pa and the measured cosmological constant $\Lambda \approx 1.1 \times 10^{-52}$ m⁻² yield:

$$G \approx \frac{(1.1 \times 10^{-52})(3 \times 10^8)^4}{8\pi(5.3 \times 10^{-10})} \approx 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}. \quad (5)$$

The significance of this result lies in the interpretation of G as an emergent "stiffness" of the dark-matter field. By utilizing global cosmological parameters (u_ν, Λ), we mathematically show that the physical "push back" from the vacuum condensate equates to the observed strength of gravitational attraction.

4.4 Novel Correlation and Derivation

It is important to emphasize that while the mathematical components utilized in this derivation—the Gross–Pitaevskii equation, the heat kernel expansion for induced gravity, and the Friedmann cosmological parameters *et al.*—are established in their respective fields, the **correlation** presented here is unique. We demonstrate that these independent frameworks are not merely parallel but are coupled through a specific mechanical-to-geometric matching condition. By showing that the mechanical bulk modulus of a dark-matter medium is identically equal to the EH prefactor, we provide a new physical derivation for the value of G , transforming it from a bare fundamental constant into a calculable property of the vacuum condensate.

While the GPE describes the medium and the Heat Kernel describes the geometry, standard physics keeps them separate. The contribution pertains to the Stiffness Matching Principle. We treat the Einstein-Hilbert action as the 'stress-strain' equation for the vacuum. By equating the mechanical resistance of the dark-matter condensate to the geometric resistance of spacetime, G is no longer a 'given' parameter; it becomes the specific value required for the medium and geometry to remain in equilibrium."

5 Dimensional Consistency and Physical Interpretation

To ensure the mathematical validity of the proposed mechanical route, we perform a dimensional analysis of the identity $G = \Lambda c^4 / (8\pi u_\nu)$. As shown in Table 1, the units of the derived coupling perfectly match the standard dimensions of Newton's gravitational constant.

Table 1: Dimensional Analysis of the Mechanical Route to G

Quantity	Symbol	SI Units	Dimensions
Gravitational Constant	G	$\text{m}^3 \text{kg}^{-1} \text{s}^{-2}$	$[L]^3 [M]^{-1} [T]^{-2}$
Speed of Light	c	m s^{-1}	$[L] [T]^{-1}$
Vacuum Energy Density	u_ν	$\text{kg m}^{-1} \text{s}^{-2} \text{ (Pa)}$	$[M] [L]^{-1} [T]^{-2}$
Cosmological Constant	Λ	m^{-2}	$[L]^{-2}$
Condensate Density	ρ_m	kg m^{-3}	$[M] [L]^{-3}$

Table 2: Algebraic Breakdown of Dimensional Consistency

Component	Dimensional Operation
Numerator ($[\Lambda] \times [c]^4$)	$[L]^{-2} \times ([L][T]^{-1})^4 = [L]^2 [T]^{-4}$
Denominator ($[u_\nu]$)	$[M][L]^{-1}[T]^{-2}$
Final Ratio ($\frac{\text{Numerator}}{\text{Denominator}}$)	$\frac{[L]^2 [T]^{-4}}{[M][L]^{-1}[T]^{-2}} = [\text{L}]^3 [\text{M}]^{-1} [\text{T}]^{-2}$
Identity Check	$\equiv [G]$

Physical Significance of the Result

The dimensional alignment supports the following physical interpretations of the theory:

- **Energy Density as Field Pressure:** The vacuum energy density u_ν (measured in Pascals) represents the physical "push back" of the dark-matter condensate against gravitational compression.
- **Curvature as Strain:** The cosmological constant Λ (measured in inverse area) functions as the intrinsic strain or geometric deformation of the medium over cosmological scales.
- **Gravity as an Emergent Stiffness:** The gravitational constant G is the resulting modulus that scales the relationship between energy density and geometry, functioning as the emergent "stiffness constant" of the universal vacuum.

6 Evolution across Cosmological Epochs

The mechanical interpretation of G implies a relationship with the condensate density ρ_m that may vary across cosmological time. In a tracking regime where ρ_m follows the background density evolution, the gravitational coupling G scales inversely with the field density.

7 Phenomenological Implications

The variable nature of G in this framework offers a potential resolution to several contemporary cosmological tensions. Notably, the high-redshift massive galaxies recently observed by the

Table 3: Evolution of G and ρ_m (Hypothetical Tracking Regime)

Epoch	Redshift (z)	Rel. Density ρ_m/ρ_0	Rel. Coupling G/G_0
Radiation Dom.	> 3000	$\sim 10^{10}$	$\sim 10^{-10}$
Matter Dom.	~ 1	8	0.125
Present Day	0	1	1

James Webb Space Telescope (JWST) suggest a faster growth of structure than predicted by standard Λ CDM.¹² A stronger gravitational coupling G in the early universe—driven by a lower initial condensate density ρ_m in certain models—may provide the necessary acceleration for such early assembly.

Furthermore, the mechanical interpretation of G impacts black hole physics; in a denser, "stiffer" vacuum (higher ρ_m , lower G), Hawking radiation temperatures $T_H \propto (GM)^{-1}$ would be suppressed, altering evaporation rates. Finally, this Condensate Link Framework may address challenges in Fuzzy Dark Matter (FDM) by linking gravitational strength to the local field pressure, potentially resolving the "cusp-core" problem by softening the potential where the condensate density is highest.

8 Conclusion

In this work, we have demonstrated that Newton's gravitational constant G can be derived from the mechanical bulk modulus of a dark-matter vacuum condensate. By identifying the vacuum as a barotropic fluid with energy density u_ν , we matched the hydrodynamic field pressure to the geometric stiffness of the Einstein–Hilbert action. This mechanical route not only recovers the observed value of G through cosmological observables but may provide a dynamic framework that accounts for anomalies in early structure formation observed by JWST and structural tensions in dark matter halos. Our results suggest that gravity is an emergent phenomenon, with G functioning as the physical stiffness of the universal medium.

Declarations

AI Transparency Statement

The fundamental theoretical framework, the novel correlation of matching mechanical bulk modulus to geometric stiffness, and the conceptual derivation of G are the original work of the author. Generative AI (Google Gemini 3 Flash) was utilized specifically for LaTeX document formatting, bibliography organization, research and assistance with mathematical formulas, notation and scientific prose to meet journal standards.

Acknowledgments

8.1 Novel Correlation and Synthesis of Established Frameworks

The derivation presented in this work is a synthesis of several established pillars of modern physics. We acknowledge and express profound gratitude to the foundational contributions of the scientific community including, but not limited to; the quantum condensate community, particularly the work on the Gross–Pitaevskii equation, the pioneers of analogue gravity, and the teams responsible for the Planck and JWST missions. Their pursuit of precision and their willingness to explore the boundaries of physics provided the data and theories required to anchor this mechanical route. Furthermore, this derivation relies on the heat kernel methods

developed within the quantum field theory community and the high-precision measurements of cosmological parameters provided by the global astrophysical community.

The unique contribution of this research is the **correlation** of these disparate frameworks through a specific mechanical-to-geometric matching condition. By demonstrating that the bulk modulus of the dark-matter medium is identically equal to the Einstein–Hilbert prefactor, we provide a new physical derivation for the value of G . This transforms G from an empirical constant into a calculable property, honoring the insights of these various fields while providing a unified mechanical explanation for the strength of gravity.

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors, and remains an independent contribution to the field.

A One-loop derivation of the EH term

We integrate out the field fluctuations ϕ . The divergent part of the one-loop effective action Γ_1 is given by the Schwinger-proper time representation: [9,10](#)

$$\Gamma_1 = -\frac{1}{2(4\pi)^2} \int d^4x \sqrt{-g} \int_{1/\Lambda_{UV}^2}^{\infty} \frac{ds}{s^3} e^{-sM_{\text{eff}}^2} [a_0 + a_1 s + a_2 s^2]. \quad (6)$$

The term $a_1 = (\frac{1}{6} - \xi)R$ induces the EH action:

$$S_{\text{ind}} \supset \frac{(\frac{1}{6} - \xi)\Lambda_{UV}^2}{32\pi^2} \int d^4x \sqrt{-g} R. \quad (7)$$

Matching this to $S_{\text{EH}} = \frac{c^4}{16\pi G} \int \sqrt{-g} R$ anchors G to the medium’s cutoff Λ_{UV} .

References

- [1] W. G. Unruh, Phys. Rev. Lett. **46**, 1351 (1981).
- [2] M. Visser, Class. Quantum Grav. **15**, 1767 (1998).
- [3] C. Barceló, S. Liberati, and M. Visser, Living Rev. Relativ. **14**, 3 (2011).
- [4] A. Einstein, Annalen der Physik **354**, 769 (1916).
- [5] D. Hilbert, Konigl. Gesell. d. Wiss. Göttingen, Nachr. Math.-Phys. Kl. 395 (1915).
- [6] E. P. Gross, Nuovo Cimento **20**, 454 (1961).
- [7] L. P. Pitaevskii, Sov. Phys. JETP **13**, 451 (1961).
- [8] L. D. Landau and E. M. Lifshitz, *Fluid Mechanics*, 2nd ed. (Pergamon, London, 1987).
- [9] D. V. Vassilevich, Phys. Rep. **388**, 279 (2003).
- [10] J. Schwinger, Phys. Rev. **82**, 664 (1951).
- [11] O. Lahav and A. R. Liddle, in *Review of Particle Physics*, PTEP **2022**, 083C01 (2022).
- [12] M. Boylan-Kolchin, Nature Astron. **7**, 731 (2023).